

1 Are the statements correct

1.a This is false

$A \neq A^*$ we can proof this by counterexample, let $A = \{1\}$ then A^* equals the regular expression 1^* , which corresponds to every possible combination of 1 including ε , and it is obvious that $1 \neq 111\dots$

1.b This is true

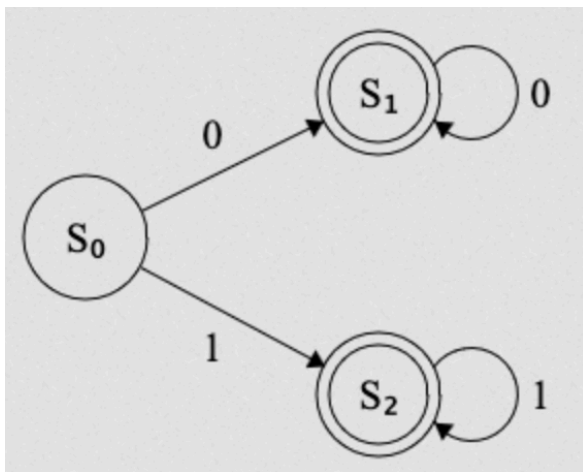
Since A^* includes all combinations of A it also includes A so it does not matter whether or not it is concatenated

1.c This is true

The order of operations does not matter for concatenation

2 State diagrams

2.a



2.b

Here is the code for the NFA tester

Init: q_0

Accept: q_1, q_2

$q_0, 0, q_1$

$q_0, 1, q_2$

$q_1, 0, q_1$

$q_2, 1, q_2$

2.c

To do the same task as a DFA we would need to add a trap state to the NFA that catches the inputs not currently in the NFA. S_1 would then transition into itself on a 0 but into S_3 on a 1 and S_2 would transition into itself on a 1 but into S_t on a 0. S_t would transition into itself on both 0 and 1. So we would need 4 states in total, one more than in the NFA.

3 Funnis application language

3.a NFA

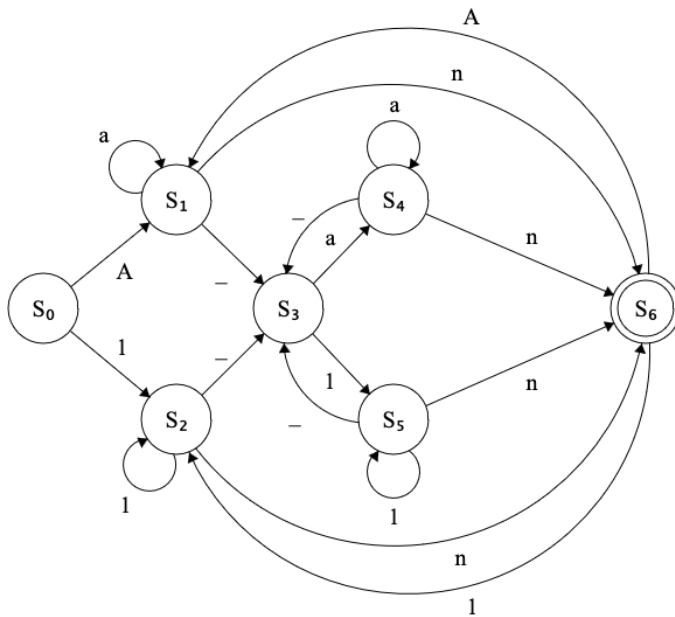


Figure 1: Here is a NFA for the language, this shows that the language is in fact regular

3.b Simulator

code

Init: q0

Accept: q6

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q0,A,q1
q0,1,q2
q1,a,q1
q1,n,q6
q1,s,q3
q2,1,q2
q2,n,q6
q2,s,q3
q3,a,q4
q3,1,q5
q4,a,q4
q4,s,q3
q4,n,q6
q5,1,q4
q5,s,q3
q5,n,q6
q6,A,q1
q6,1,q2
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